

Basic principles of plyometric training

Robert Heiduk

Introduction

Plyometric training is commonly performed with the aim of improving power, movement, speed and/or economy of movement. The aim of this chapter is to provide information about its origins and to review potential physiological mechanisms. In addition, this chapter provides some guidelines and recent developments for application and training program design to optimize performance associated with plyometric training.

Origin of plyometric training

The term plyometric is composed of two Greek words: plio, meaning “more”, and metric, meaning “to measure”. Measurable increase might be the most accurate definition (Chu, 1998). Commonly known as plyometrics, this category of exercise is mostly referred to as jump training. The earliest references have been found in a publication by Zanon (1989). In the early 1960s, Soviet Union track and field coach Yuri Verkhoshansky experimented with maximal jumps and hops to increase power in his high level athletes. He found that “depth” jumping, with landing and take-off on both feet, was particularly effective for improving jump performance. He termed his new training the “Shock Method”. The name referred to the mechanical shock stimulation used to force the muscle to produce as much tension as possible (Verkhoshansky and Siff, 2009). Verkhoshansky’s strange new system of jumps

and bounds quickly became very popular in the world of sport. In Germany it was popularized by Peter Tschiene, in South Africa by Mel Siff, and in Italy by Carmello Bosco (Verkhoshansky and Siff, 2009). It has been suggested that the dominance of the Eastern European countries in track and field, weightlifting and gymnastics during the 1970s can be partially attributed to the Shock Method (Chu, 1998). In the USA, the term “plyometrics” was coined in 1975 by Verkhoshansky’s colleague, the track and field coach Fred Wilt (Chu, 1998).

The stretch-shortening cycle

Walking, running, throwing, jumping, hopping, swinging, hitting and kicking are examples of activities that allow muscle-tendon units (MTUs) to be stretched before shortening rapidly, without delay, between the eccentric and concentric phases. This is known as the stretch-shortening cycle (SSC) (Komi, 2000). There are variations in time characteristics in different types of SSC movement patterns. Güllich and Schmidtbleicher (2000) classify SSC movements into a short (<200 ms) and long (>200 ms) SSC. Examples of a short SSC are sprinting with a ground contact time of ~100–110 ms, the long jump with 120 ms and the high jump with 170–180 ms (Bührlé, 1989). The long SSC can be observed in the jump of a volleyball smash with ground contact times of 300–360 ms.

Compared with the pure concentric rate of force development, without a preceding

Chapter 25

eccentric movement phase, use of the SSC creates a significant performance enhancement (Komi, 2000). For example, Cavagna (1964) demonstrated an increase in running economy of ~50%, thus maximizing energy efficiency and improving aerobic performance (Spurrs et al., 2003). In plyometric patterns like bouncing, Dean and Kuo (2011) observed how the properties of the elastic tissues reduce the work needed from active muscle fibers, which leads to a decrease in energy expenditure. In maximum performance, Markovic (2004) found an 8% gain in vertical jump height by utilizing the SSC.

Physiological mechanisms

Currently, elastic and contractile mechanisms are considered to be responsible for performance-enhancing effects in the SSC (Blazevich, 2011). The relative contribution of each mechanism seems to be dependent on the force and time characteristics of the movements. In addition, individual anthropometrics and genetics come into play (Sano et al., 2015). The energy storage and spring-like release of potential energy in elastic tissues, like tendons, mean that MTUs shorten at speeds exceeding those using muscle contraction on its own (Blazevich, 2011).

Zatsiorsky and Kraemer (2008) point out that the amount of elastic energy storage in tendons is largely dependent on the stiffness of the muscles. If we consider muscle and tendon as two springs connected in series, more energy is stored on the compliant spring. Muscle stiffness is dependent on the level of contractile force a muscle can produce, which depends on factors such as muscle size and neural activation.

An additional factor for the importance of muscle force during a countermovement is the ability to decelerate in the eccentric or yielding phase. This specific contractile ability seems to

be a supporting factor for the recoil mechanisms in elastic tissues.

Kawakami et al. (2002) showed that, during the eccentric phase of a countermovement plantar flexion, the gastrocnemius only works in a near-isometric manner, which means that only very few length changes in the muscle fibers are observable in this phase. Because, by contrast, the tendon elongates significantly, the SSC cannot be considered as a muscle-related phenomenon. Kangaroos are a perfect example of an efficient use of the SSC (Morgan et al., 1978). Despite comparably small muscles, their long and compliant hind limb tendons allow the storage of large amounts of kinetic energy, which can be used to achieve tremendously long jumping distances. While the role of the tendon in the SSC has been extensively researched, to date there are no data for the deep fascia. However, a similar contribution is conceivable, albeit probably to a smaller degree. In summary, the higher power output observed during the SSC can, at least partially, be attributed to the recoil properties of the connective tissue. A second key factor in the SSC is the amount of coupling time between the eccentric and concentric phase, which is described as the “amortization phase” (Chu, 1998). If the amortization phase lasts too long, the energy being absorbed is lost as heat (Radcliffe and Farentinos, 1999). Schmidtbleicher and Gollhofer (1985) found that in trained athletes the amortization phase was ~100 ms, significantly shorter than in untrained subjects. Other research suggests that the ideal coupling time lasts 15–25 ms (Bosco et al., 1981). Chu (1998) characterizes the amortization phase as quasi-isometric muscle action, which means that muscle length does not significantly change during this period.

Besides elastic recoil and eccentric-concentric contraction coupling, another mechanism for

Basic principles of plyometric training

enhanced performance in SSC movements suggests that the rapid pre-stretch of the muscle during the eccentric phase leads to a stretch reflex, which is initiated in the muscle spindle receptor. Consequently, a more powerful muscle contraction is produced, due to the activation of more motor units (Komi, 1992). Rassier and Herzog (2005) propose that the magnitude of force enhancement may come from three different factors: (1) the magnitude of stretch, (2) the rate of stretch and (3) the duration of stretch. The faster the eccentric loading phase then the stronger the concentric muscle contraction (Böhm et al., 2006). Bubeck (2002) adds, that under different loading conditions, the neuromuscular activation patterns change. This would indicate that adaptations in a SSC are highly context-specific. Some authors suggest that the stretch reflex may be not be initiated in all muscles (Nardone et al., 1990): monoarticular muscles, which cross one joint only, may benefit more strongly from the stretch reflex than biarticular muscles (Nicol and Komi, 1998).

The spring-like properties of the elastic tissues vary for subjects of different genders, injury state and athletic background. This would be one explanation for the differences in the effects of plyometric training. It seems that there are movement- and age-specific differences in the behavior of elastic tissues in plyometric training (Arampatzis et al., 2007; Hoffrén et al., 2007).

Preconditions for plyometric training

The implementation of proper plyometric training requires some preliminary considerations. The success of training programs for optimal individual adaptations strongly relies on personalization. At the beginning a specific goal should be defined. Like every other kind of exercise, the training effects are highly movement-specific.

A plyometric training program for golfers is different to one for sprinters, jumpers, gymnasts, strength athletes, or for rehabilitation. Moreover, the current performance level of the athlete has to be considered. A beginner needs a different training stimulus to an intermediate or a top-level athlete. Age, body weight, skill and previous injury play an important role (Holcomb et al., 1998). Starting with plyometric training, the development of a proper exercise technique under qualified instruction and supervision is paramount. The goal is to increase efficiency and avoid injury.

Injury prevention is a critical factor in plyometric training. High impulsive peak forces produce serious injury risks for the joints. Coaches should be aware that plyometric exercises can lead to muscle damage and collagen breakdown of the connective tissues without decreases in skeletal muscle capacity (Tofas et al., 2008). Therefore, risks may result from inappropriate or excessive use, rather than from plyometric training itself (Verkhoshanky and Siff, 2009). Being technically proficient and creating appropriate progression seem to be the most important factors in injury prevention. In addition, the coach has to ensure a proper warm-up and the use of an adequate landing surface (Borkowski, 1990). Wathen (1994) highlights that the quality of the landing surfaces plays a major role in performing plyometric drills. Allerheiligen and Rogers (1996) suggest spring-loaded floors or rubber mats and caution against practicing on concrete or asphalt. Firm natural grass may also be a sufficient training surface. However, overly soft surfaces diminish the effectiveness of plyometrics, as the mechanical stimulus on the body is lowered.

It is suggested that a carefully designed plyometric training regime, over the course of a season, may reduce the incidence of knee injury

Chapter 25

in athletes involved in sports with large jumping and cutting components (Hewett et al., 1999). In addition to the enhancement of the athlete's movement skills, the increased musculotendinous stiffness may be another possible factor for injury prevention (Spurrs et al., 2003).

Depth jumps are the most advertized form of plyometric training, but they also exert the highest loads on body structures (Allerheiligen and Rogers, 1996). Thus, these should only be used by a small percentage of athletes. The jump height is the most critical factor. People weighing more than 98 kg should not perform depth jumps from platforms higher than 45 cm (Chu, 1992). Ishikawa et al. (2005) reported that different drop heights in depth jumps have a specific impact on connective tissues. It seems that overly high eccentric loads limit the effectiveness of the catapult mechanism (Figure 25.1).

Strength and conditioning practitioners should never design a plyometric training routine as a stand-alone program. Every athlete needs an adequate strength base prior to



Figure 25.1

Incorporating depth jumps require a well-founded strength base and precise technique. These should only be used by athletes who have advanced to levels that are appropriate for them.

With permission from Pullsh, Germany.

starting plyometric drills. When strength is not adequate, it results in a loss of stability and forces are excessively absorbed by the joints. This leads to an increased risk of injury. For high-intensity drills, like depth jumps, the literature suggests a strength ratio of 1.5–2.5 times body weight in the One Repetition Maximum (1-RM) barbell-squat (Allerheiligen and Rogers, 1996; Chu, 1992; Gambetta, 1992).

These recommendations may not be appropriate for female athletes and, to date, no specific research has addressed this topic. Experience in daily sport practice suggests that for female athletes 1–1.5 times body weight in the 1-RM barbell-squat would be a suitable value. There is also a lack of research on recommendations for strength values in upper body plyometrics and for application in recreational athletes.

In addition, a sufficient amount of specific flexibility is needed to perform plyometric training safely (Chu, 1992). Davies and Matheson (2001) report the following contraindications to plyometrics: pain, inflammation, ligament and capsular sprains, muscle and tendon strains, joint instability, connective tissue limitations based on postoperative conditions, and a lack of strength base.

Basic principles for training program design

First and foremost, there is no general design that can be applied consistently to all competitive athletes. Hence, at this juncture, it would make no sense to lay out a specific training regime because the requirements in each area of training are highly individualized. However, there are some basic rules that apply.

Two different kinds of guidelines are useful when creating a training program. The first

Basic principles of plyometric training

approach uses coaching principles, which describe instructions in a deeper context, such as educational aspects to ensure optimal training progression. The four coaching principles in this approach are:

1. Move from simple to complex
2. Move from light to heavy
3. Technique first – quality over quantity
4. Avoid fatigue.

These principles suggest systematically increasing loads and the complexity of movements. Moreover, the principles promote the importance of movement quality instead of quantity—the coach has to ensure the perfect execution of the movement and must avoid fatigue. In the SSC, fatigue is absolutely counterproductive. It causes a decrease in neuromuscular activity and a loss of elastic energy potential (Komi, 2000). This leads to reduced muscle and joint stiffness, which can cause injury. Thus, turning plyometrics into a conditioning program, which is usually designed to develop energy systems, is not advisable. One key to avoid fatigue is to keep a specific work:rest ratio between sets. For rehabilitation, Chu and Cordier (2001) suggest a ratio of 1:5–1:10. When the exercise takes 10 s, 50–100 s rest would be appropriate. True Shock Training, for example, depth jumps, demands considerably longer rest periods. Depending on the intensity, 3–10 min rest is advised (Bubeck, 2001; Sialis, 2004).

The second approach determines the specific load. In plyometric training, intensity is determined by the jumping height or effort, the usage of additional weight, or the style of exercise, like single- or double-leg exercises. Volume and frequency describe more quantitative variables, such as the amount of work within a training session, the total number of jumps, the number of

sets and repetitions, the work:rest ratio between exercise and break within a training session, and the number of sessions per week. According to the literature, a sufficient recovery time of 48–72 h is required. This means a training frequency of approximately twice a week (Chu and Cordier, 2001).

Fluid periodisation

From a biological point of view, general training frequency suggestions can be far removed from the individual's needs, because time courses of adaptation are never linear but instead are based on instability. Therefore, recovery times are not always equal, not even in the same athlete. This is due to environmental changes, as well as asynchronous reactions, of different biological systems in the body.

As previously stated, true plyometric training is very taxing on the nervous system, therefore, for optimal trainability, the timing of a plyometric training session has to be scheduled when the central nervous system (CNS) is in a state of optimal readiness without fatigue. Measurements of the brain's electrical direct current (DC) potential have been proven to be a valid and reliable method for assessment of the functional state of the CNS (Morris, 2015). DC potential is an ultraslow (<0.5 Hz) brainwave that is sensitive to short- and long term adaptational changes in the organism (Iliukhina, 2011).

Practically spoken, measuring DC potential is valuable for detecting CNS fatigue as a result of the effects of different training sessions and interventions. With this additional information it is possible to make adjustments to individual adaptation, because the coach has more information to work with and, via long-term monitoring trends, it can be deduced how an individual responds to certain forms of training. In summary, gaining

Chapter 25



Figure 25.2

Measurement of the brain's electrical direct current (DC) potential with a mobile computer device to detect adaptational changes in the organism and determine the current trainability of the athlete. The setup consists of one electrode on the forehead, a reference electrode on the palm for the central nervous system (CNS), and a chest unit for measuring the heart's electrical activity at the same time.

With permission from Omegawave, Finland

a more objective insight enables optimization of training beyond just guessing or general recommendations (Figure 25.2).

Eccentric overload training

The modality of eccentric overload training has a reasonable value as part of a comprehensive training program, particularly in regard to the importance of specific strength ability in the deceleration phase of the SSC and related MTU stiffness. An eccentric muscle contraction occurs when the external force exceeds the force developed by the muscle.

In a systematic approach to eccentric training, there has to be distinguishment between training with eccentric emphasis and eccentric overload. The first can be employed with almost all resistance training exercises, because this modality is simply a conventional strength training with a

slower execution in the yielding phase, that is, 1 s concentric and 3 s eccentric contraction.

This eccentric training modality has different effects that are not specific to the physiological changes seen in the SSC or eccentric overload. Eccentric emphasis training can lead to significant hormonal reactions and elevations in resting metabolic rate, immediately after a training session (Hackney et al., 2008; Preuß, 2010). Nevertheless, in a long-term approach, this modality has value for the body's preparation before progressing to higher loads, as well as providing a means to support recovery during de-load periods.

A true eccentric overload is achieved when the external force is on a level where an isometric contraction is almost impossible. With regard to the SSC a specific eccentric overload has to go beyond submaximal slow eccentrics. This is because, in the eccentric phase, higher loads than the voluntary concentric maximum are possible (Güllich and Schmidtbleicher, 2000), and in the braking phase movement speeds are quite high. In traditional strength training, which is based on gravity, respectively, mass, this is called supramaximal load, (i.e. 105% of the 1-RM).

For a safe and effective eccentric overload, gravity-based equipment, like barbells, dumbbells or kettlebells, are less convenient, because these tools expose the body to an unnecessarily high risk of injury. Spotters are also always needed to keep the weight moving and to overcome the weaker concentric phase.

Therefore, the implementation of eccentric overload strength training has been lacking in traditional strength and conditioning program designs. In addition, traditional weight training using gravity-dependent equipment is limited to vertical action in the frontal plane, leaving out

Basic principles of plyometric training

most horizontal and lateral actions in the mode of eccentric overload.

Flywheel training has been proven to close the gap where traditional gravity-based weights and machines have their limits. Instead of working against gravity, in flywheel training the resistance is created through the inertia of a flywheel, which is accelerated or decelerated with muscle force. The level of inertia of the flywheel rather than the weight determines the force needed. The principle is equivalent to that of a traditional yoyo device. Thanks to technological progress and growing research on eccentric training, flywheel training is enjoying a remarkable comeback (Figure 25.3).

The principle of the flywheel became popular in the late 1980s when researchers began to focus on how to maintain lean muscle mass and bone density during extended space travel (Dudley et al., 1991). The effectiveness of flywheel training for developing strength, hypertrophy, power, injury prevention and rehabilitation has been confirmed by extensive research (Alfredson, 2003; Gonzalo-Skok et al., 2016). For improvements in

plyometric performance, it is noteworthy that recent research indicates that using eccentric overload training leads to a significantly greater braking and propulsive contact time (de Hoyo et al., 2016). From a practical point of view, the simplicity of training and infinite amount of exercise variation within an eccentrically overloaded environment is the biggest advantage of flywheel training. Last but not least, rehabilitation training with flywheel devices has proven to be an effective means for tendinopathy (Murtaugh and Ihm, 2013), and for the elderly (Onambélé et al., 2008).

Wearable resistance

With increasing performance levels of athletes, specificity of training is paramount to ensure a transfer to performance, because neuromuscular and functional changes induced by exercise are specific to the mode of exercise performed. Thus, plyometric training must exactly match the movement kinetics, kinematics, and metabolic demands of the sport. Therefore, the sport itself would be the most specific way to improve

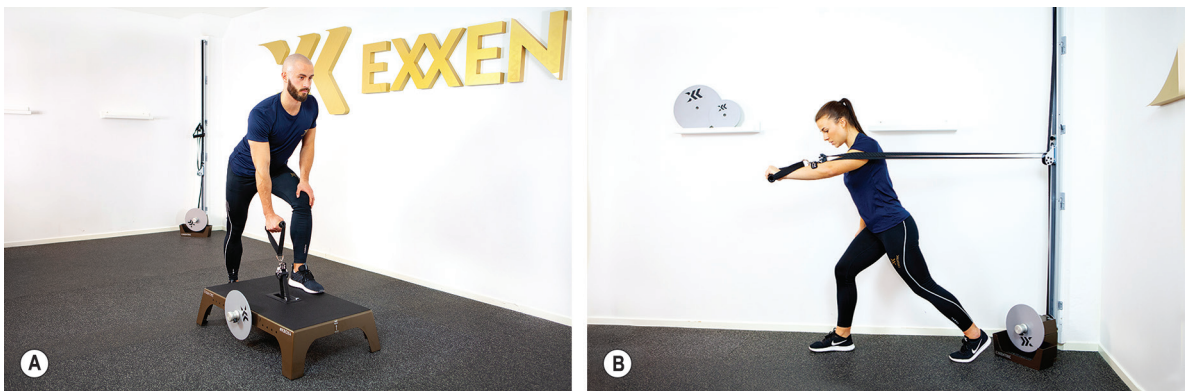


Figure 25.3

A flywheel device is a safe and effective way to apply an eccentric overload at high movement speeds in any movement pattern.

With permission from Exxentric, Sweden.

Chapter 25

performance. (The most specific plyometric training for a sprinter is sprinting.)

Unfortunately, in traditional resistance training, the equipment defines the movement, not the user. This means that the user is attached to a piece of equipment and the characteristics of the equipment determine what he or she can do with it.

The approach of wearable resistance is exactly the opposite by matching resistance equipment to movement. This is achieved by new garment technology, where flexible drop-shaped weights of 50, 100 or 200 g can be stuck to increase the weight of a particular body part. The protocols and loading conditions are different to traditional resistance training. Applied loads are generally measured and progressed in grams not in kilograms and total loading generally does not exceed 10% of body weight in order to maintain movement speeds within 10% of competition speed. Furthermore, the loads should not disrupt the athlete's skill-based range of motion. Here, there is greater focus on maintaining the body-mind connection in a proprioceptively enriched resistance training because in these protocols there is less emphasis on traditional resistance parameters, like repetitions or weight (Figure 25.4).

Conclusions

Reviewing the development of plyometric training from its origins to the present, few significant changes are observable since the 1980s. However, decisive progress has been made in complementary and overarching methods, mainly based on the further development of new technologies. These technological advances open up new opportunities to increase efficiency of training, gaining results with less cost of adaptation and providing a more specific, demand-oriented approach.

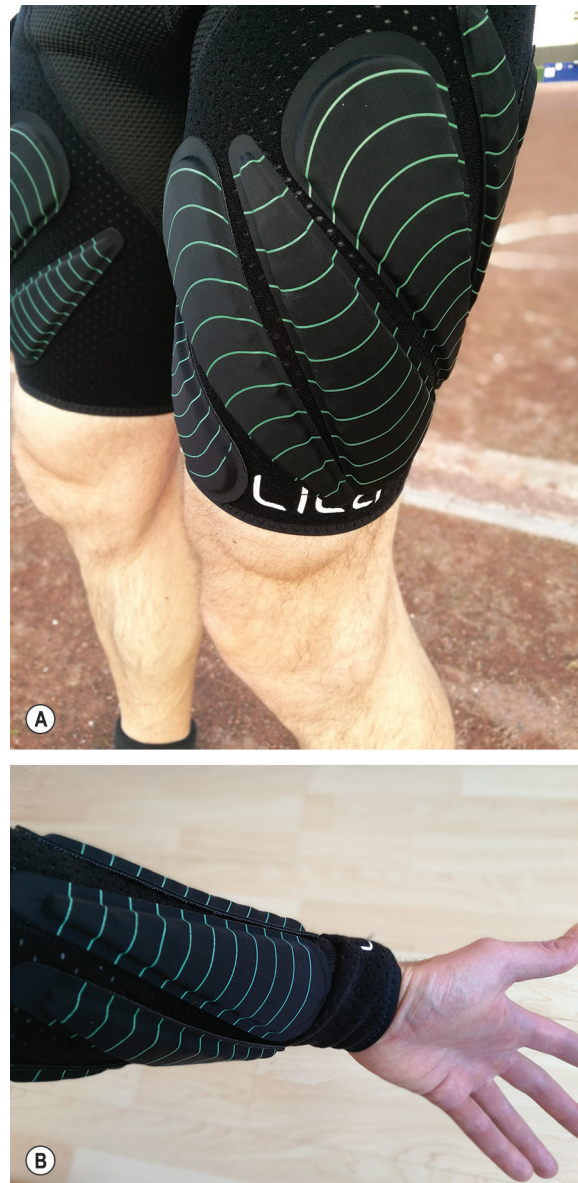


Figure 25.4

Wearable resistance on different body parts. The flexible drop-shaped weights (50, 100 and 200 g) can be placed at any body part in any direction. This methodology allows a proprioceptively enriched resistance training, since the flexible weights are not perceived as foreign bodies, as in conventional weight vests.

With permission from Robert Heiduk, Germany.

Basic principles of plyometric training

References

- Alfredson, H. (2003) Chronic midportion achilles tendinopathy: An update on research and treatment. *Clin Sports Med.* 22: 727–741.
- Allerheiligen, B. & Rogers, R. (1996) Plyometrics program design. National Strength and Conditioning Association (Eds), *Plyometric and Medicine Ball Training*. Colorado Springs: National Strength and Conditioning Association, 3–8.
- Arampatzis, A., Karamanidis, K., Morey-Klapsing, G., De Monte, G. & Staflidis, S. (2007) Mechanical properties of the triceps surae tendon and aponeurosis in relation to intensity of sport activity. *J Biomech.* 40: 1946–1952.
- Blazevich, A. (2011) The stretch-shortening cycle (SSC). In: Cardinale, M., Newton, R. & Nosaka, K. (Eds), *Strength and conditioning: Biological principles and practical applications*. Chichester, UK: Wiley-Blackwell, 209–222.
- Böhm, H., Cole, G., Brüggemann, G. & Cole, H. (2006) Contribution of muscle series elasticity to maximum performance in drop jumping. *J Appl Biomech.* 22: 3–13.
- Borkowski, J. (1990) Prevention of pre-season muscle soreness: plyometric exercise. Abstract. *Athletic Training*. 25: 122.
- Bosco, C., Komi, P.V. & Ito A. (1981) Pre-stretch potentiation of human skeletal muscle during ballistic movement. *Acta Physiologica Scandinavica*. 111: 135–140.
- Bubeck, D. (2002) Belastungsvariation und funktionelle Anpassungen im Dehnungs-Verkürzungs-Zyklus. PhD Thesis. Fakultät für Geschichts-, Sozial- und Wirtschaftswissenschaften der Universität Stuttgart.
- Bührle M. (1989) Maximalkraft-Schnellkraft-Reaktivkraft. *Sportwissenschaft*. 3: 311–325.
- Chu, D. (1992) *Jumping into Plyometrics*. Champaign, IL: Human Kinetics.
- Chu, D. & Cordier D. (2000) Plyometrics in Rehabilitation. In: Ellenbecker T.S. (Ed), *Knee Ligament Rehabilitation*. New York, NY: Churchill Livingstone, 321–344.
- Davies, G.J. & Matheson, J.W. (2001) Shoulder plyometrics. *Sports Med Arthrosc.* 9: 1–18.
- Dean, J.C. & Kuo, A.D. (2011) Energetic costs of producing muscle work and force in a cyclical human bouncing task. *J Appl Physiol.* 110: 873–880.
- de Hoyo, M., Sanudo, B., Carrasco, L., Mateo-Cortes, J., Dominguez-Cobo, S., Fernandes, O. & Gonzalo-Skok, O. (2016) Effects of 10-week eccentric overload training on kinetic parameters during change of direction in football players. *J Sports Sci.* 34: 1380–1387.
- Dudley, G.A., Tesch, P.A., Miller, B.J. & Buchanan, P. (1991) Importance of eccentric actions in performance adaptations to resistance training. *Aviat Space Environ Med.* 62: 543–550.
- Gambetta, V. (1992) New plyometric training techniques: designing a more effective plyometric training program. *Coaching Volleyball*. April/May: 26–28.
- Gonzalo-Skok, O., Tous-Fajardo, J., Valero-Campo, C., Berzosa, C., Bataller, A.V., Arjol-Serrano, J.L., Mendez-Villanueva, A. (2016) Eccentric overload training in team-sports functional performance: constant bilateral vertical vs. variable unilateral multidirectional movements. *Int J Sports Physiol Perform.* 1–23.
- Güllich, A. & Schmidtbleicher, D. (2000) Struktur der Kraftfähigkeiten und ihrer Trainingsmethoden. In: Siewers, M. (Ed.), *Muskelkrafttraining. Band 1: Ausgewählte Themen – Alter, Dehnung, Ernährung, Methodik*. Kiel: Siewers Eigenverlag, 17–71.
- Hackney, K.J., Engels, H.J. & Gretebeck, R.J. (2008) Resting energy expenditure and delayed-onset muscle soreness after full-body resistance training with an eccentric concentration. *J Strength Cond Res.* 22: 1602–1609.
- Hewett, T.E., Lindenfeld, T.N., Riccobene, J.V. & Noyes, F.R. (1999) The effect of neuromuscular training on the incidence of knee injury in female athletes. *Am J Sports Med.* 27: 699–705.
- Hoffrén, M., Ishikawa, M. & Komi, P.V. (2007) Age-related neuromuscular function during drop jumps. *J Appl Physiol.* 103: 276–283.
- Iliukhina, V.A. (2011) Continuity and prospects of development of researches in area system-integrativity of psychophysiology of functional state and cognitive activity. *Human Physiol.* 37: 105–123.
- Ishikawa, M., Niemälä, E. & Komi, P.V. (2005) Interaction between fascicle and tendinous tissues in short-contact stretch-shortening cycle exercise with varying eccentric intensities. *J Appl Physiol.* 99: 217–223.
- Komi, P.V. (1992) Stretch shortening cycle. In: *Strength and Power in Sport*.

Chapter 25

- Oxford, UK: Blackwell Scientific, 169–179.
- Komi, P.V. (2000) Stretch-shortening cycle: a powerful model to study normal and fatigued muscle. *J Biomech.* 33: 1197–1206.
- Morris, C.W. (2015) The Effect of Fluid Periodization on Athletic Performance Outcomes in American Football Players. PhD dissertation, University of Kentucky. Theses and Dissertations – Kinesiology and Health Promotion. 24.
- Murtaugh, B. & Ihm, J.M. (2013) Eccentric training for the treatment of tendinopathies. *Curr Sports Med Rep.* 12: 175–182.
- Nardone, A., Corra, T. & Schieppati, M. (1990) Different activations of the soleus and gastrocnemii muscles in response to various types of stance perturbation in man. *Exp Brain Res.* 80: 323–332.
- Nicol, C. & Komi, P.V. (1998) Significance of passively induced stretch reflexes on Achilles tendon force enhancement. *Muscle Nerve J.* 21: 1546–1548.
- Onambélé, G.L., Maganaris, C.N., Mian, O.S., Tam, E., Rejc, E., McEwan, I.M. & Narici, M.V. (2008) Neuromuscular and balance responses to flywheel inertial versus weight training in older persons. *J Biomech.* 41: 3133–3138.
- Preuß, P. (2010) Muskuläre Leistung im Krafttraining: Analyse verschiedener Formen der Bewegungsausführung auf die Maximal- und Schnellkraft nach der Methode der submaximalen Kontraktionen bis zur Erschöpfung unter Berücksichtigung akuter hormoneller Auslenkungen. Dissertation. Köln, Germany.
- Radcliffe, J.C. & Farentinos, B.C. (1999) *High-Powered Plyometrics*. Champaign, IL: Human Kinetics.
- Rassier, D. & Herzog, W. (2005) Force enhancement and relaxation rates after stretch of activated muscle fibres. *Proc Royal Soc B Biol Sci.* 272: 475–480.
- Sano, K., Nicol, C., Akiyama, M. et al. (2015) Can measures of muscle–tendon interaction improve our understanding of the superiority of Kenyan endurance runners? *Eur J Applied Physiol.* 115: 849.
- Schmidtbleicher, D. & Gollhofer, A. (1985) Einflussgrößen des reaktiven Bewegungsverhaltens und deren Bedeutung für die Trainingspraxis. In: Bührle, M. (Ed.), *Grundlagen des Maximal- und Schnellkrafttrainings*. Schorn-dorf: Hoffmann, 271–281.
- Sialis, J. (2004) Innervationscharakteristik und Trainingsadaptabilität im Dehnungs-Verkürzungs-Zyklus. Stuttgart, Germany: University of Stuttgart.
- Spurrs, R.W., Murphy, A.J. & Watsford, M.L. (2003) The effect of plyometric training on distance running performance. *Eur J Appl Physiol.* 89: 1–7.
- Tofas, T., Jamurtas, A.Z., Fatouros, I., Nikolaidis, M.G., Koutedakis, Y., Sinouris, E.A., Papageorgakopoulou, N. & Theocharis, D.A. (2008) Plyometric exercise increases serum indices of muscle damage and collagen breakdown. *J Strength Cond Res.* 22: 490–496.
- Verkhoshansky, Y. & Siff, M.C. (2009) *Supertraining*. Sixth edition, expanded version. Rome: Verkhoshansky SSTM.
- Wathen, D. (1994) Literature review: explosive plyometric exercises. In: National Strength and Conditioning Association (Eds), Position Paper and Literature Review: Explosive Exercises and Training and Explosive Plyometric Exercises, pp.13–16. Colorado Springs: National Strength and Conditioning Association.
- Zanon, S. (1989) Plyometrics: past and present. *New Studies in Athletics.* 4: 7–17.
- Zatsiorky, V.M. & Kraemer, W.J. (2008) *Krafttraining – Praxis und Wissenschaft*. Third Edition. Aachen: Meyer & Meyer.